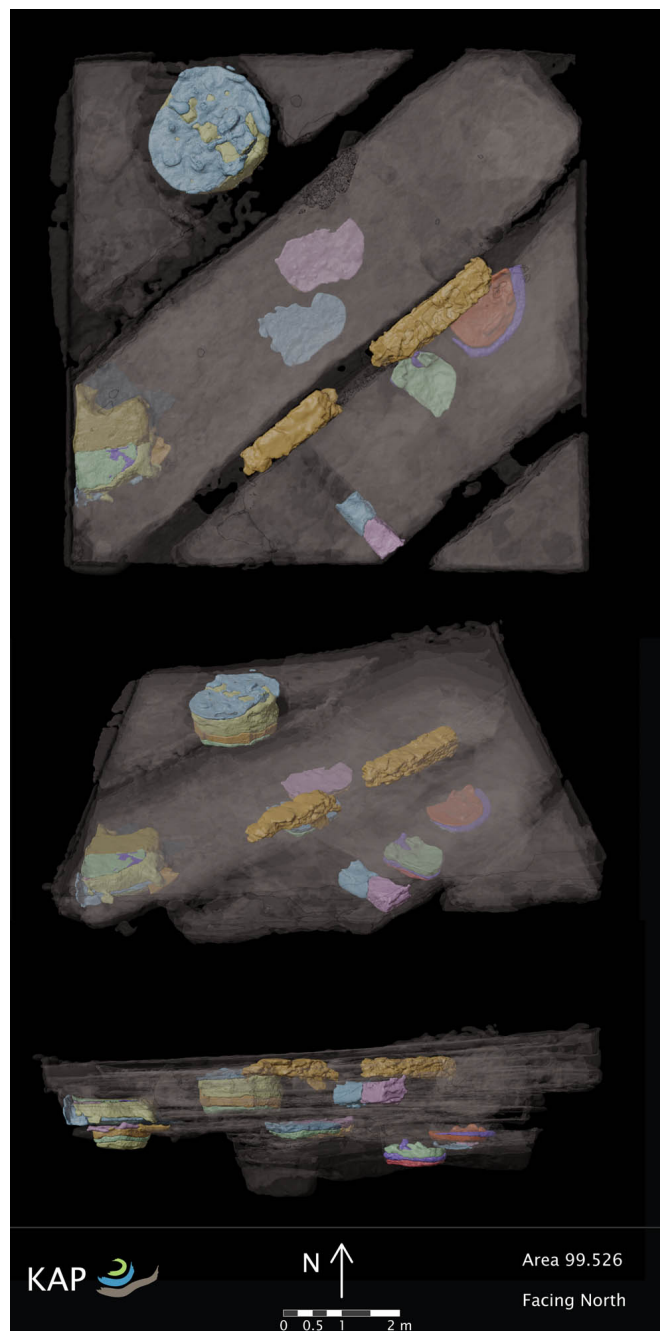


Excavation

Archaeological **excavation** is a primary method by which we learn about societies of the past. By digging carefully and systematically, and recording what we find as we find it, archaeologists can identify the **stratigraphy** of a site: the various superimposed layers of human activity and natural sediments that have buried the site. Archaeologists can date these **strata**, or layers, relative to one another; absent unusual circumstances, deposits or structures that lie below others are earlier, and those that cover others are more recent. Using careful photogrammetric recording of the material removed and GPS recording of key points, it is possible to render excavated material in a virtual three-dimensional space, and to manipulate the resulting shapes and spaces digitally.





Photograph (top) and 3-D rendered photogrammetric model (in three views: top-down, oblique, section) of one excavation area at Kaymakçı. Colored shapes are archaeological strata that were removed during excavation.

Source: [Open Archaeology](https://www.degruyter.com/document/doi/10.1515/opar-2020-0149/html).

(<https://www.degruyter.com/document/doi/10.1515/opar-2020-0149/html>)

Excavation is the practice of digging carefully and systematically to uncover features, artifacts, and strata.

Stratigraphy is study of layers of rock (in geology), sediments, and soils (in geology and archaeology), and the relative location of features and artifacts (in archaeology). Based on the relative position of layers, features, and artifacts, it is possible to determine the chronological sequence of their deposition.

Strata are distinct layers of soils or sediments discovered during excavation.

Sampling Food Residues

At Kaymakçı, one of the team's goals was to determine what plant and animal foods people ate. We designed a sampling strategy so that when, during excavation, we encountered deposits of sediment that we believed dated to the period in which people lived at the site, we sampled those for plant and animal remains. In future modules, you will learn more about the sampling methods used at Kaymakçı and other sites, but for now, we will discuss only the samples taken to find seeds preserved by being burned into inorganic carbon.

Paleoethnobotanical samples consist of bags of excavated sediment, with the location from which they were collected carefully recorded. We use a system of water **flotation** to separate the sediment from the seeds, which are collected, dried out, and then examined using a microscope to determine from which plants they came. The seeds are counted and then analyzed in various ways to paint a picture of people's diet in the past and the association of plants with specific spaces—for example, food storage or cooking spaces.





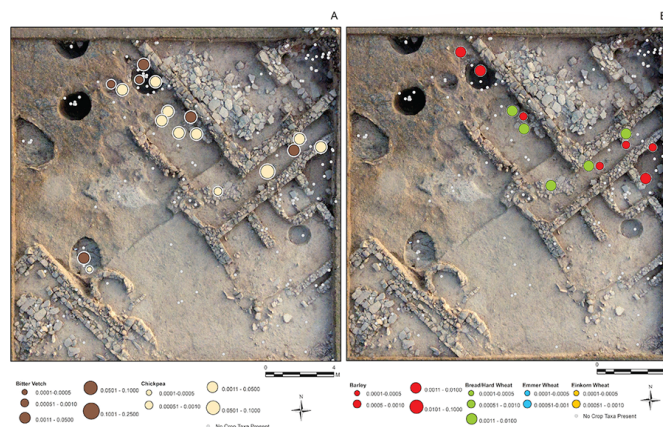
Nami Shin (CAS '15) conducting flotation at Kaymakçı.

Source: Dr. Marston



Nami Shin (CAS '15) at the microscope sorting botanical remains from Kaymakçı at the BU Environmental Archaeology Laboratory.

Source: BU Today (<https://www.bu.edu/today/close-ups/2013/studying-ancient-seeds/>)



Spatial analysis of the locations of pulses (small-seeded legumes) and cereal grains in one excavation area.

Flotation is a process by which archaeologists use water to separate plant material, usually carbonized wood and seeds, from soil and heavy artifacts and ecofacts, such as pottery shards and animal-bone fragments. This is the primary method by which paleoethnobotanical remains are collected in most areas of the world.